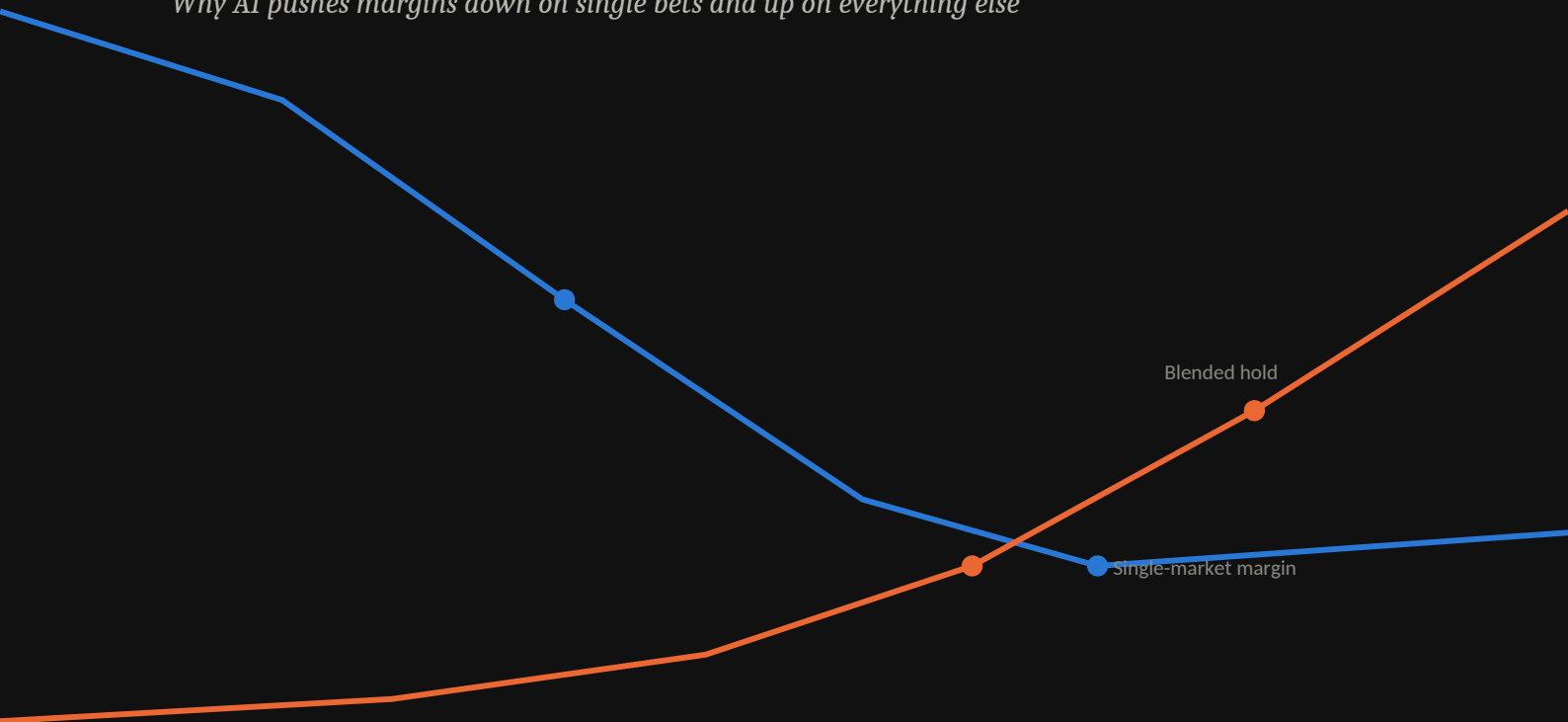


The Margin Paradox

How Artificial Intelligence Is Reshaping Bookmaker Margins in Sports Betting

Why AI pushes margins down on single bets and up on everything else



Dennis Buchbauer

Predictive & Betting Analytics

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ABSTRACT

Is AI making betting cheaper or more expensive?

Over the last two decades, the price of a football bet has fallen dramatically. The bookmaker margin (overround) on a Premier League home-draw-away market has dropped from roughly 10% in the late 1990s to around 4–5% today. At the same time, the share of every dollar wagered that sportsbooks keep as revenue (the *hold*) has risen in every single year since the US market opened, from about 7.0% in 2019 to about 10.2% in 2025. Both statements are true at once. This paper explains why.

I argue that artificial intelligence and machine-learning-based pricing are the common cause of both trends. Better forecasting reduces the bookmaker's risk of being picked off by informed bettors, which allows thinner margins on liquid, transparent markets. The same technology makes it possible, for the first time, to price correlated, multi-leg and in-play products at scale – bet builders, same-game parlays, player props and micro-bets – whose fair price the bettor cannot compute, and whose margins run between 15% and over 40%. Combined with AI-driven customer profiling that limits winning bettors, the result is what I call **the Margin Paradox**: headline odds look sharper than ever, while the average bettor pays more than ever.

The paper synthesises more than 50 academic studies, regulatory datasets and company filings, and adds original calculations: a decomposition of New Jersey hold into single and parlay components, a compounding model for multi-leg margins, and an estimate of the effective cost of a bet in the German regulated market under the 5.3% betting tax.

≈10% → 4.65%

Premier League 1X2 overround, late 1990s vs. 2024/25

7.0% → 10.2%

US national sportsbook hold, 2019 vs. 2025

32% → 73%

Parlays in New Jersey: share of handle vs. share of revenue (Sept 2024)

Key findings

- 1. Two-speed market.** Margins on single, liquid markets (1X2, handicaps, totals) fell by more than half between 2000 and 2018 and have stabilised at 4–5% for the average bookmaker and around 3% at Pinnacle.
- 2. The reversal has started.** Since 2020/21, the average Premier League 1X2 margin has crept back up from 4.12% to 4.65%, and the Bundesliga from 4.72% to 5.02%. Compression is over.
- 3. Product mix is the new margin lever.** Parlays held 17.5% in New Jersey (2024) and 21.9% in Illinois (2025), versus single-bet hold of roughly 4–5%. AI pricing is what makes these products possible.
- 4. Winners are filtered out.** In Great Britain, 46.8% of restricted customers were in profit versus 25.4% of all customers (2024). AI risk-profiling protects the margin on the bettor side.
- 5. Bettor-side AI rarely wins long-term.** Closing odds remain the best public forecast. Profitable models exist in research, but they depend on calibration rather than accuracy and are quickly limited.
- 6. Germany pays twice.** The 5.3% stake tax pushes the effective cost of a Premier League 1X2 bet in Germany to around 9.5%, double the underlying market margin, and the DSWV puts the average German payout ratio at 85%.

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About the author

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SECTION 1

1. Introduction: a market rebuilt by prediction

Sports betting is, at its core, a forecasting business. A bookmaker sells a probability with a mark-up attached; a bettor buys it when they believe the probability is wrong. For most of the industry's history, that probability was set by human traders with spreadsheets, gut feeling and a phone line to rival shops. Today, the price of almost every market – from the Champions League final to the next corner kick in a Bulgarian second-division match – is generated, monitored and adjusted by machine-learning models.

The scale of this shift is easiest to see through the suppliers. Sportradar now covers more than one million matches a year [36], and its AI pricing product *Alpha Odds* is deployed across 80% of the events its managed trading service handles [37]. Kambi rebuilt its trading around an AI division and today describes its soccer pricing as “largely powered by our AI trading division” [35]. DraftKings paid \$134.6 million for Simplebet, a company whose only product is machine-learning pricing for in-play micro-markets [39].

The obvious question for bettors, operators and regulators is what this does to the *price* of a bet. If bookmakers forecast better, should they not be able to offer tighter odds? Or does better forecasting simply allow them to extract more from customers? The honest answer, supported by the data in this paper, is: **both – depending on which product you look at.**

Research questions

- How have bookmaker margins on standard football markets developed from the late 1990s to 2025?
- How has the realised margin (hold) of the whole market developed in the AI era, and what drives it?
- What role do AI-priced complex products, in-play markets and AI-driven customer profiling play?
- Can bettors use AI to beat the market, and what does the academic evidence say?
- What are the specific consequences for the German regulated market?

Definitions used in this paper

Table 1. Core concepts.

Term	Definition
Overround / margin	The amount by which the implied probabilities of all outcomes exceed 100%. Odds of 2.00 / 3.40 / 3.80 imply $50.0\% + 29.4\% + 26.3\% = 105.7\%$, i.e. a 5.7% overround.
Hold	Realised gross gaming revenue divided by handle (total stakes). Hold reflects margin, product mix and results luck.
Structural hold	The hold an operator expects to earn over time given its product mix, before results variance. Used by DraftKings and others in investor reporting.
Payout ratio	The share of stakes returned to bettors (100% minus hold). In German: <i>Auszahlungsquote</i> .
Parlay / SGP / bet builder	A combination bet that wins only if all legs win. A same-game parlay (SGP) or bet builder combines legs from one match, which are statistically correlated.
Closing line	The last odds before an event starts; widely regarded as the most efficient public forecast.

SECTION 2

2. Why forecasting quality should change the price

Economic theory gives a precise reason why the quality of a bookmaker's forecasts should affect its margin. In Hyun Song Shin's classic model [19], a bookmaker faces a mix of uninformed bettors and a small share of insiders who know more than the odds reveal. The overround is, in effect, an insurance premium against those insiders: the bookmaker must charge everyone enough to cover losses to the few who are better informed. Because insiders do the most damage on longshots, the margin is loaded onto unlikely outcomes – one of the classic explanations for the favourite–longshot bias.

The implication is direct. **When the bookmaker's own forecasts improve, the information gap shrinks, the adverse-selection risk falls, and the required insurance premium – the margin – can come down.** An AI model that ingests line-ups, tracking data, weather, market moves at 50 other bookmakers and late team news in milliseconds closes exactly the gap that insiders used to exploit.

But theory also offers a second, less comfortable insight. Steven Levitt's study of an NFL betting contest [20] showed that bookmakers are better forecasters than bettors, and that they do not simply balance their books. Instead, they set prices strategically to exploit predictable bettor biases – and this earned them more than a balanced book would. Kuypers [21] showed that a profit-maximising bookmaker can rationally set inefficient odds. Better forecasting, in other words, does not only reduce risk; it also improves the bookmaker's ability to identify where bettors misprice outcomes and to charge accordingly.

These two mechanisms – lower adverse-selection risk and better exploitation of bettor misperception – pull margins in opposite directions. Which one dominates depends on how transparent a market is. On a Premier League 1X2 market, every bettor can compare prices across 50 bookmakers in seconds; competition forces the risk saving to be passed on. On a four-leg same-game parlay, no bettor can compute the fair price, and no comparison site can display it. That is where the second mechanism takes over.

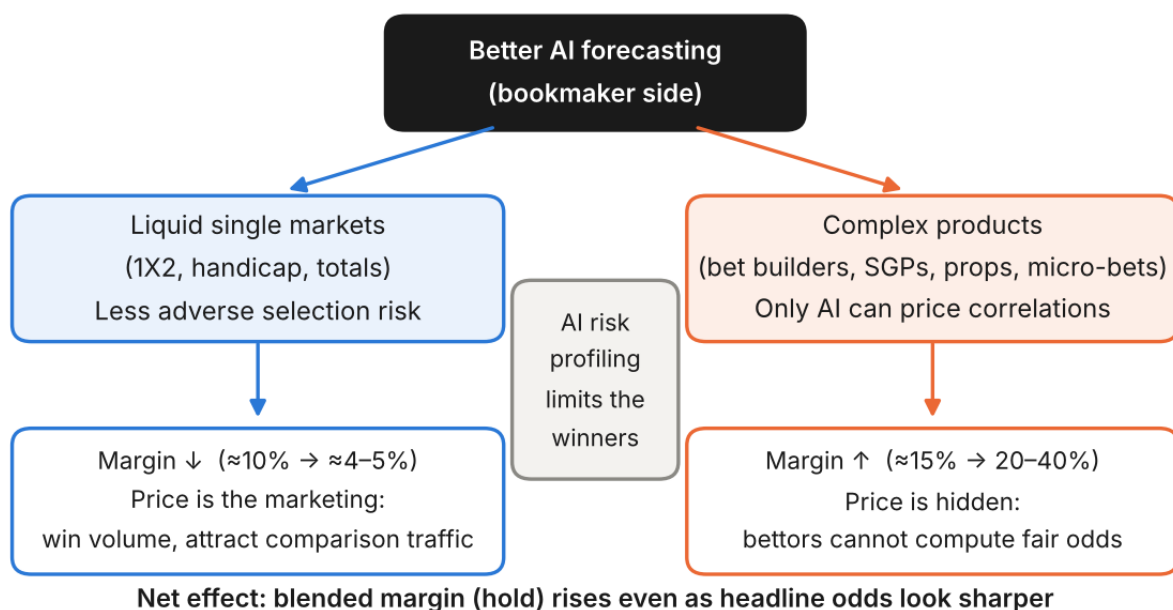


Figure 1. The Margin Paradox framework. Better AI forecasting compresses margins where prices are transparent and expands them where prices are opaque, while AI risk profiling limits the bettors who could exploit either. Source: author's framework.

SECTION 3

3. Phase one: margin compression on single markets

The long-run decline of football betting margins is one of the best-documented facts in sports economics. In the late 1990s, UK bookmakers worked with a near-constant margin of around 10.5% on home/draw/away bets [25]. Gomez-Gonzalez and del Corral, analysing 2000/01 to 2016/17, describe a “continuous decrease in the prices of the odds, especially noticeable in the top five leagues”, accompanied by a growing number of cross-bookmaker arbitrage opportunities [14].

For the Premier League specifically, a study of six large bookmakers reports an average margin of about 9% in 2005/06 falling to about 4% by 2017/18; for the Championship to League Two the decline was from 11% to 6% [24]. Reade, Singleton and Vaughan Williams measured a mean Premier League overround of 4.8% across 51 online bookmakers for 2010/11 onward [22]. Whelan and Hegarty, using 151,683 European matches from 2005/06 to 2024/25, find that both overround-implied and realised loss rates have fallen over the past decade, “perhaps reflecting greater competition” [17].

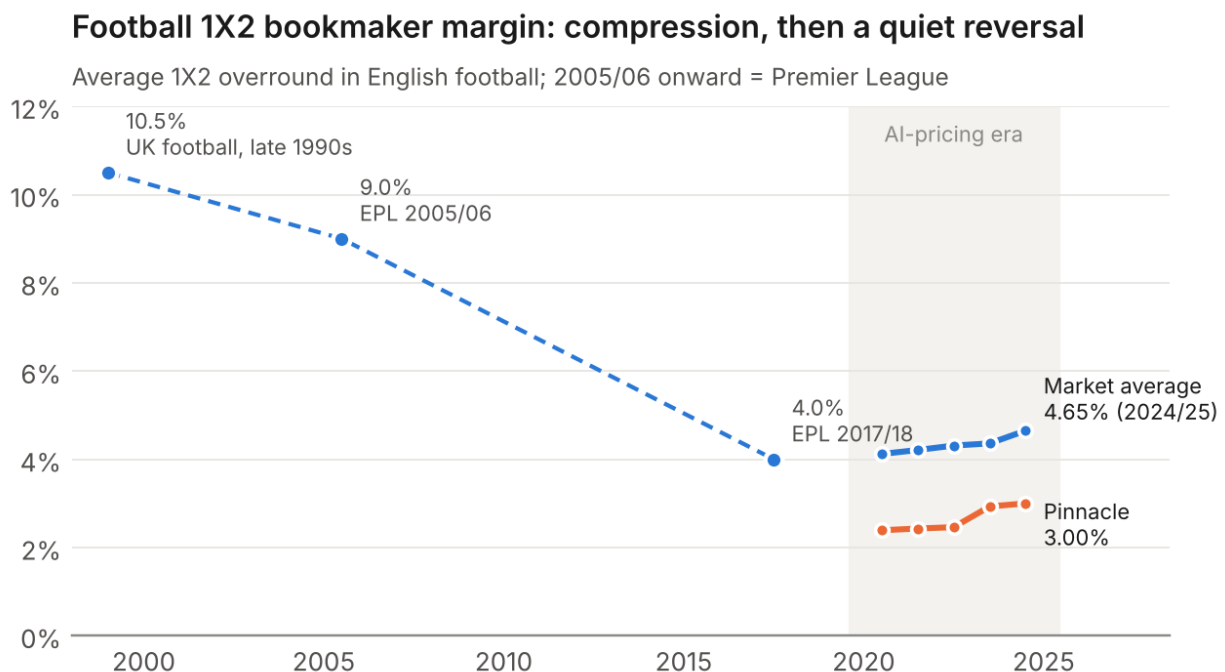


Figure 2. Long-run development of the 1X2 margin in English football. Historic points come from different studies and bookmaker samples and are connected with a dashed line to signal that; 2020/21–2024/25 values are a consistent series. Sources: Newall (Forrest & Simmons; Kuypers) [25], six-bookmaker study as summarised in [24], Eastwood/pena.lt [23].

What drove the compression?

It would be wrong to attribute the first phase of margin compression mainly to AI. Che, Feddersen and Humphreys show that the fall in Premier League overrounds at the two largest UK bookmakers is linked to the arrival of the Betfair exchange, not simply to the growth of online bookmakers [15]. Odds-comparison websites, the low-margin Asian model represented by Pinnacle, and intense customer-acquisition competition did the heavy lifting between 2000 and 2015.

What AI changed was the *sustainability* of thin margins. A 3–4% margin leaves little room for pricing errors. Forrest, Goddard and Simmons already observed that bookmakers’ forecasts became “increasingly effective” as competition intensified, because the financial penalty for imprecise odds grew [1]. Today’s odds

are remarkably efficient: Elaad, Reade and Singleton find that odds across 51 online bookmakers are “generally not biased” against actual outcomes [4]; Angelini and De Angelis find 8 of 11 European leagues efficient at the best available odds [3]; and Winkelmann et al. show that inefficiencies over 14 seasons are “not persistent or systematic across leagues” [5]. In-play prices, too, update “swiftly and fully” to goals [7].

Table 2. Selected evidence on the efficiency of football betting odds.

Study	Sample	Key finding on efficiency
Forrest, Goddard & Simmons (2005) [1]	≈10,000 English matches	Bookmaker forecasts improved over 5 years; statistical models fail to outperform them
Spann & Skiera (2009) [8]	Bundesliga, 3 seasons	Odds and prediction markets beat tipsters; 25% state margin made every strategy unprofitable
Angelini & De Angelis (2019) [3]	41 bookmakers, 11 leagues, 11 years	8 of 11 markets efficient at best odds
Elaad, Reade & Singleton (2020) [4]	51 bookmakers, 16,000+ matches	No systematic bias in aggregate; individual books under-use rivals’ information
Winkelmann et al. (2024) [5]	14 seasons, European leagues	Single-season anomalies are not persistent
Hegarty & Whelan (2025) [16]	≈150,000 matches	Asian handicap: efficient, lower margin; 1X2: strong favourite–longshot bias

The quiet reversal since 2020

The most important – and least discussed – finding for the AI era is that compression has stopped. An analysis of roughly 250 million betting lines by Martin Eastwood [23] shows the average Premier League 1X2 overround rising in every season from 4.12% (2020/21) to 4.65% (2024/25). The Bundesliga moved from 4.72% to 5.02%. Even Pinnacle, the industry’s low-margin benchmark, widened from 2.39% to 3.00%.

Table 3. 1X2 overround by season, 2020/21–2024/25. Source: Eastwood (2025) [23].

Season	Premier League avg.	Bundesliga avg.	Pinnacle (EPL)
2020/21	4.12%	4.72%	2.39%
2021/22	4.21%	4.63%	2.43%
2022/23	4.31%	4.71%	2.46%
2023/24	4.36%	4.76%	2.93%
2024/25	4.65%	5.02%	3.00%
Change	+0.53 pp	+0.30 pp	+0.61 pp

My interpretation is that the competitive logic has changed. When the margin on a Premier League match was the main product, it was also the main battlefield. Today, single-match 1X2 bets are increasingly a shop window: they bring the customer in, while the money is made on bet builders and in-play markets. And when operators can identify and limit sharp bettors with AI profiling (Section 5), they no longer need to defend against them with ever-thinner prices. The pressure that drove margins down for two decades is weakening.

SECTION 4

4. Phase two: AI-priced products and rising hold

If margins on single markets fell, why is the industry earning more per euro staked than ever? The answer is in the product mix. In the United States – the best-documented market thanks to monthly state reporting – national hold has risen every year since legalisation: 7.0% in 2019, 7.5% in 2021, 8.0% in 2022, 9.1% in 2023, 9.3% in 2024 and about 10.2% in 2025 [29][30]. Revenue grew 22.8% in 2025 on handle growth of only 11.0% [29].

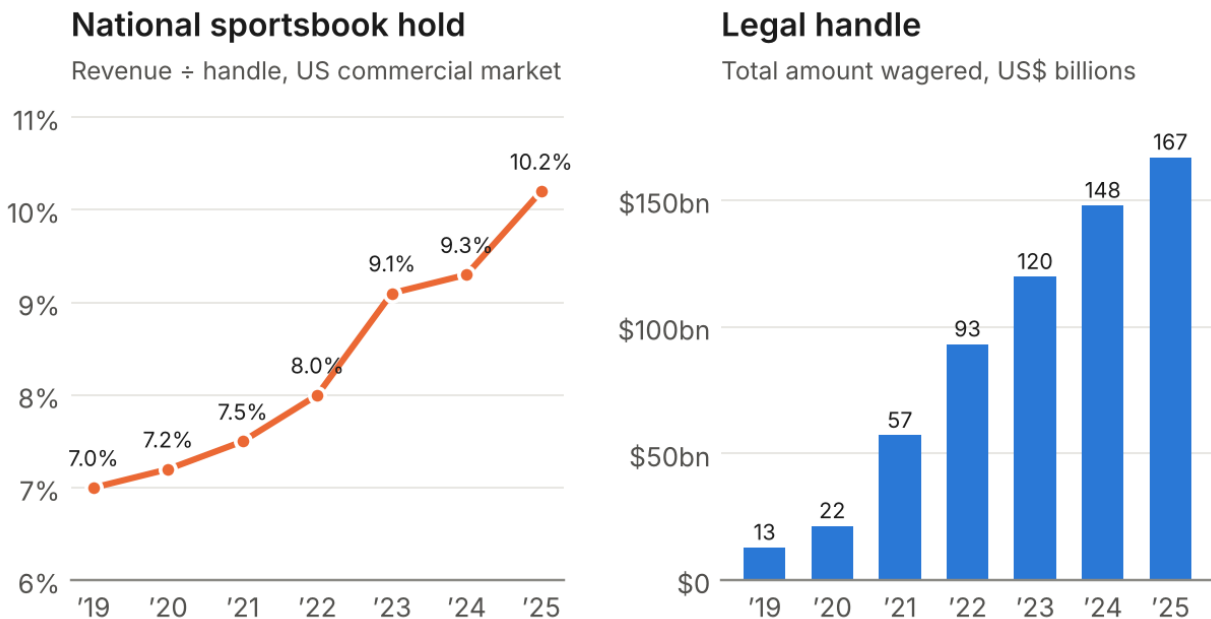


Figure 3. US commercial sports betting: hold (left) and handle (right), 2019–2025. Hold is revenue ÷ handle. Sources: American Gaming Association via SBC Americas, iGB and ESPN [29]; Legal Sports Report [30].

Parlays: one-third of the money, three-quarters of the profit

The driver is the parlay, and above all the same-game parlay. In September 2024, parlays accounted for 32.2% of New Jersey handle but 72.5% of gross revenue [18]. For the full year 2024, New Jersey parlay revenue was \$640.9 million at a hold of 17.5% [31]. In Illinois, parlays held 21.9% in calendar year 2025, while single bets held 3.9% in May 2025 [32]. FanDuel's Illinois parlay hold reached 25.4% in April 2024 [32].

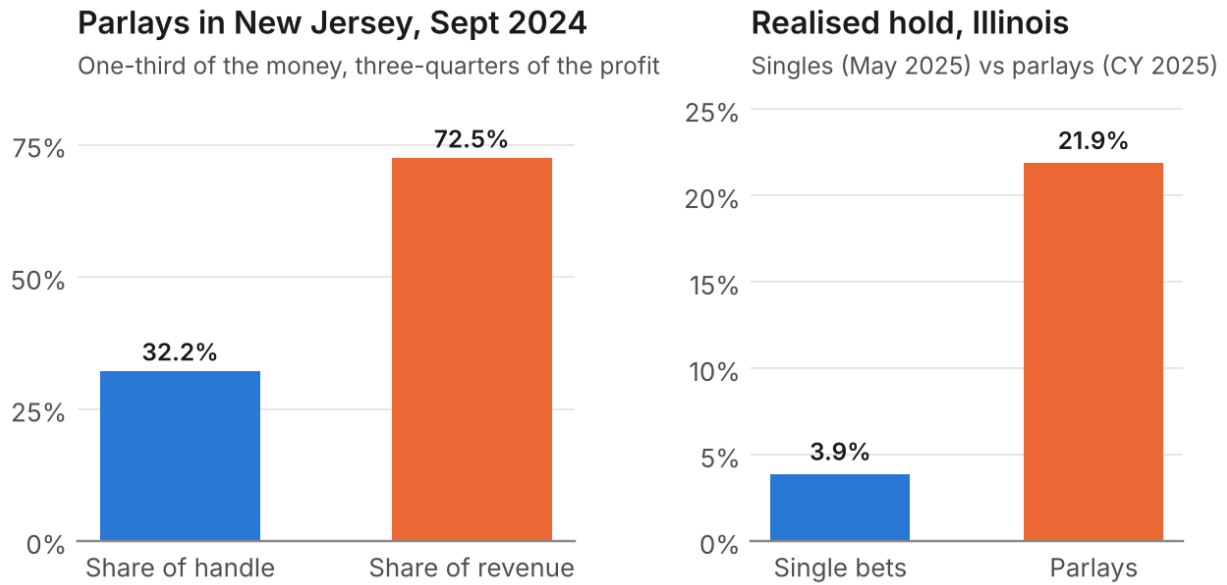


Figure 4. Left: parlay share of handle vs. share of gross revenue, New Jersey, September 2024. Right: realised hold on single bets vs. parlays in Illinois. Sources: Whelan (2026) citing state data [18]; Illinois Gaming Board via InGame/Action Network [32].

Original analysis: decomposing New Jersey's 2024 hold

New Jersey reported total sports wagering revenue of \$1.09 bn on handle of about \$12.8 bn in 2024, a blended hold of 8.5%. Parlay revenue was \$640.9 m at 17.5% hold [31]. That implies parlay handle of about \$3.66 bn, or 28.6% of the total.

Removing parlays leaves about \$9.14 bn of non-parlay handle earning about \$0.45 bn – an implied hold of **≈4.9% on everything else**. In other words: the non-parlay book in New Jersey still earns a margin close to a classic single-bet market. The entire rise of US hold above the historical ~7% level is explained by the parlay share.

The mathematics of compounding

Why are parlays so profitable? Because margins compound. If each leg carries a 5% overround, the bettor receives roughly 95.2% of fair value per leg. A two-leg parlay returns 90.7%; a five-leg parlay only 78.4%. Whelan reports that the average US parlay has about five legs and that the share of stakes lost on parlays is about four times higher than on single bets [18]. The five-leg value of 21.6% in Table 4 matches Illinois' realised parlay hold of 21.9% almost exactly.

Table 4. Compounding of per-leg margin in independent parlays (author's calculation). Same-game parlays add a further correlation premium on top.

Legs	Payout ratio (5% per leg)	Effective house edge	Payout ratio (3% per leg)	Effective house edge
1	95.2%	4.8%	97.1%	2.9%
2	90.7%	9.3%	94.3%	5.7%
3	86.4%	13.6%	91.5%	8.5%
4	82.3%	17.7%	88.8%	11.2%
5	78.4%	21.6%	86.3%	13.7%
6	74.6%	25.4%	83.7%	16.3%

For same-game parlays, compounding is only half the story. Legs within one match are correlated – a striker scoring and his team winning are not independent events – and pricing that correlation correctly requires joint simulation of the whole match. This is precisely the capability machine learning delivered. Before it, bookmakers either refused such combinations or priced them very defensively. In the UK, bet-builder margins average 20–35% depending on the bookmaker [27]; in the US, about 15% versus about 5% on singles [28]. Kambi reports that Bet Builder already accounts for around 30% of its pre-match soccer GGR [35].

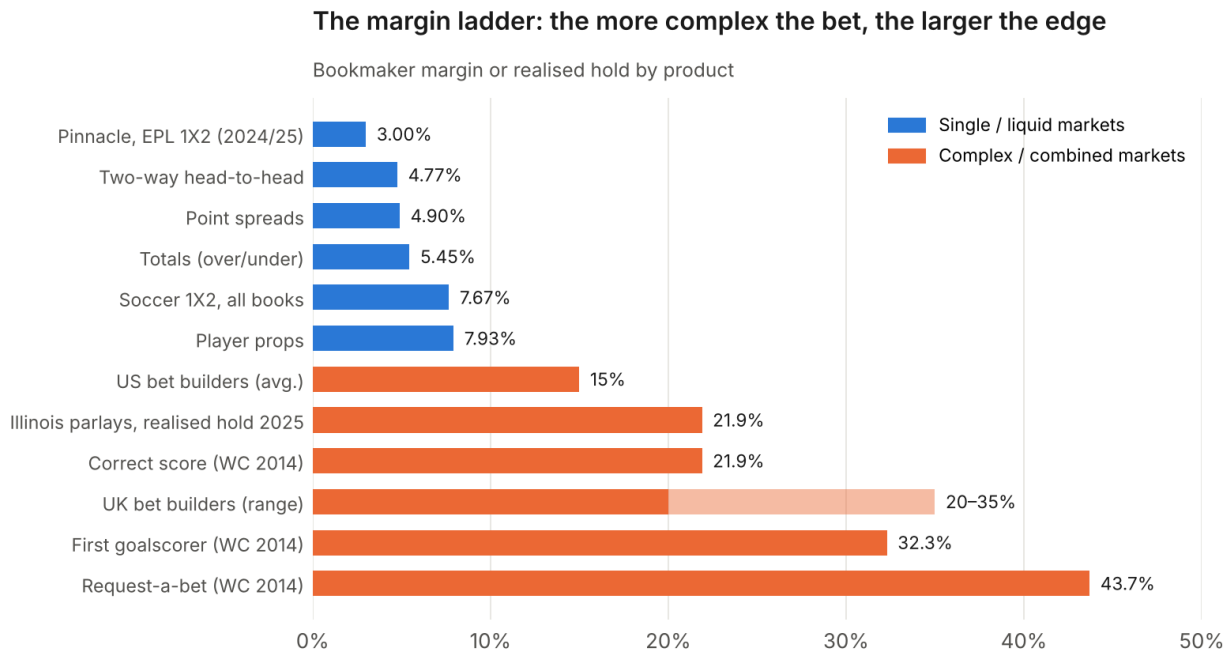


Figure 5. Margin or realised hold by product type. Sources: Eastwood [23], Glush (2026) [26], Kaplan/SBC News [28], Illinois Gaming Board [32], Newall (World Cup 2014) [25], Propus Partners/SBC News [27].

What the operators say

Operators are unusually explicit that this is technology-driven. DraftKings’ structural hold rose from about 9.5% (Q3 2023) to 10.5% (FY 2024) and about 11% (FY 2025 guidance), with parlay mix contributing +3.7 to +4.3 percentage points in 2025 quarters; its NFL 2025–26 season hold was 16% [33]. Flutter’s FanDuel lifted its gross revenue margin from 7% in 2020 to 12% in 2024 and has set a target of 16% by 2030 [34]. In Q1 2025, FanDuel attributed its 14.1% hold to “proprietary pricing technology” and same-game-parlay penetration [34]. In Europe, Kambi raised its long-term operator trading-margin expectation to 9.5–11.0%, explicitly citing bet builders and its proprietary AI, and delivered 10.8% for 2025 and 11.2% in Q4 2025 [35].

Table 5. Margin indicators reported by operators and suppliers.

Operator / supplier	Metric	Earlier	Latest	Stated driver
DraftKings [33]	Structural hold	≈9.5% (Q3 2023)	≈11% (FY 2025)	Parlay mix, pricing
FanDuel / Flutter [34]	Gross revenue margin	7% (2020)	12% (2024); target 16%	SGPs, pricing tech
Kambi [35]	Operator trading margin	10.1% (Q4 2024)	11.2% (Q4 2025)	Bet builder, AI trading
Sportradar [37]	Client profit, Alpha Odds	Conventional odds	+11% avg. (2024, vendor claim)	AI pricing

In-play and micro-betting

The third AI-enabled product is live betting. Optimove’s analysis of about 3.8 million US bettors found that 52% of wagers in 2024 were placed in-play [40]; globally, in-play accounts for an estimated \$28.4 billion of the forecast \$94 billion regulated sports betting GGR [51]. In-play markets refresh every few seconds and cannot be priced manually at scale. A study of French bookmakers on one Champions League matchday measured average overround of 14.6% on live markets versus 10.7% pre-match [26]. Micro-betting – on the next pitch, next point or next play – takes this further, and is the reason DraftKings bought Simplebet [39].

The suppliers win too

The firms supplying data and AI pricing are growing faster than the betting market itself. Genius Sports’ revenue rose 31% to \$669.5 million in 2025, citing price increases on contract renewals [38]; Sportradar grew 17% to €1.29 billion with adjusted EBITDA margin of 23% [36]. Kambi, the turnkey sportsbook supplier, shrank 8% in revenue in 2025 as large operators brought trading in-house [35] – a sign that proprietary pricing capability is becoming a strategic asset operators no longer want to rent.

Data & pricing suppliers: revenue indexed to 2023 = 100

Reported group revenue in each company's reporting currency

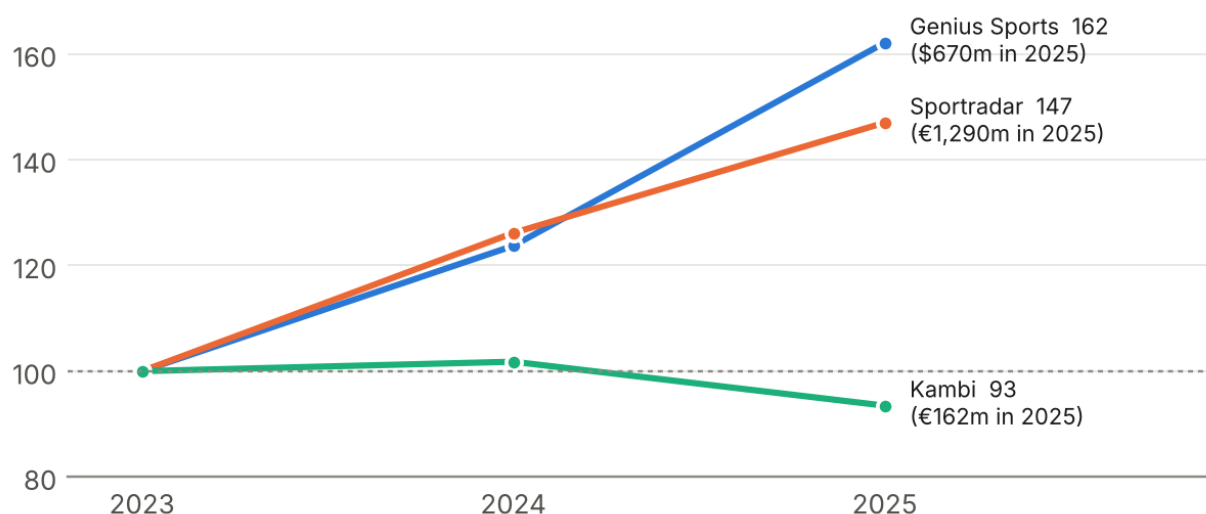


Figure 6. Revenue of major betting data and pricing suppliers, indexed to 2023 = 100. Sources: Sportradar annual results [36], Genius Sports SEC filings [38], Kambi quarterly reports [35].

SECTION 5

5. The other side of the model: who is allowed to win?

A bookmaker can tolerate thin margins only if it controls who bets into them. The same AI that prices markets is used to score customers – by stake patterns, timing, market choice, and above all by whether they consistently beat the closing line. Customers who look sharp get a reduced *stake factor*, are limited to small amounts, or are closed.

For the first time, regulators have published hard numbers. The UK Gambling Commission reported that in 2024, 643,779 of 14.9 million active accounts (4.31%) were restricted. Stake-factor reductions affected 2.68% of all active accounts. Of restricted customers, 46.8% were in profit, compared with 25.4% of all customers [41]. In Massachusetts, 0.64% of accounts were limited; the regulator found that bettors who consistently beat the closing line were more likely to be limited, while accounts whose limits were raised were more likely VIPs, and in December 2025 voted to advance a rule requiring operators to explain limits [42].

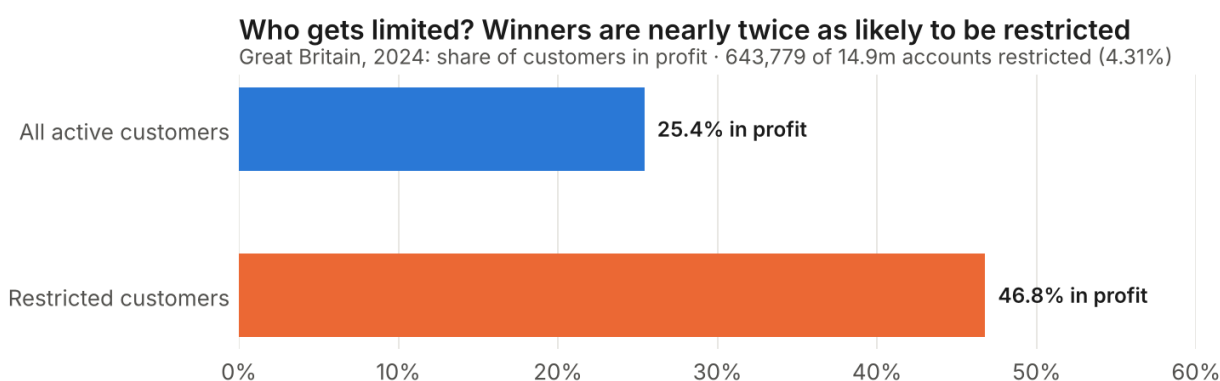


Figure 7. Share of customers in profit: restricted vs. all active customers, Great Britain, 2024. Source: UK Gambling Commission [41].

Academic researchers have experienced this first-hand. Kaunitz, Zhong and Kreiner built a strategy that used the consensus of 32 bookmakers' odds as a probability estimate and bet on outliers. It was profitable in a ten-year simulation and in real-money trading – and the bookmakers restricted their accounts within months [9]. The lesson is structural: in a market where AI identifies winners faster than winners can scale, a bettor's edge is capped not by the efficiency of prices but by the bookmaker's willingness to accept the bet.

The contrast is the low-margin, high-limit model of Pinnacle and the exchanges. Pinnacle operates on margins of around 2–3% on major football markets and states that it welcomes winning bettors, using their action to sharpen its lines [53]. Its own margin has risen from 2.39% to 3.00% since 2020/21 [23] – a reminder that even the sharpest book in the market is not immune to the broader repricing.

SECTION 6

6. Can bettors use AI to beat the market?

The democratisation of machine learning has put sophisticated tools into the hands of individual bettors. Gradient-boosted trees, neural networks and Bayesian models are available to anyone with a laptop. Does that shift the balance of power back? The research is sobering.

Bookmaker odds remain the most accurate publicly available forecast of sports results [2]. In the 2017 Soccer Prediction Challenge, the winning model by Hubáček et al. achieved a ranked probability score of 0.2063 with 52.4% accuracy [10]; in tennis, Wilkens found that neural networks and random forests could not push accuracy meaningfully above about 70% or clearly beat simple benchmarks based on odds and rankings [13].

There are, however, two important nuances. First, Hubáček, Šourek and Železný showed that a model can be profitable even when it is less accurate than the bookmaker, provided it is *decorrelated* from the bookmaker's forecast and combined with disciplined stake allocation [10]. Second, Walsh and Joshi found that choosing models by *calibration* rather than accuracy changed average betting ROI on NBA data from -35.2% to +34.7% [11]. That is a single season and should be treated with caution, but the principle is robust: for betting, a model must be right about probabilities, not just about winners.

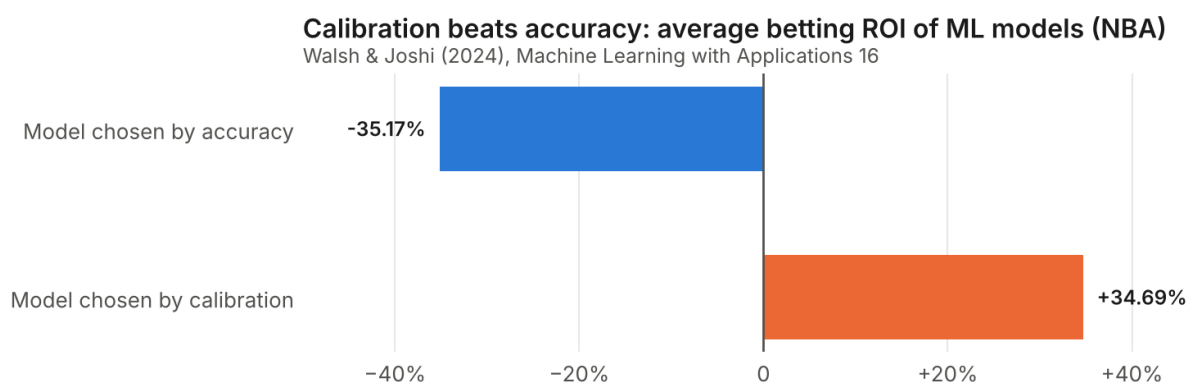


Figure 8. Average return on investment of NBA betting models selected by accuracy vs. by calibration. Source: Walsh & Joshi (2024) [11].

Wunderlich and Memmert warn that betting returns are a poor measure of forecast quality, because positive returns can arise from inaccurate models [12]. My conclusion from this literature is that bettor-side AI can find edges in niche, lower-liquidity markets and around news events – such as the temporary mispricing of home advantage after the COVID-19 break in the Bundesliga [6] – but that profitable strategies face the limitation problem described in Section 5. On complex products with 20%+ margins, no realistic model edge overcomes the price.

Table 6. Machine learning against the bookmaker: selected evidence.

Study	Approach	Result
Kaunitz et al. (2017) [9]	Consensus odds vs. outlier odds; 479,440 matches	Profitable in simulation and live; accounts limited within months
Hubáček et al. (2019) [10]	CNN on player data, decorrelated from bookmaker, portfolio staking	Positive cumulative profit on NBA 2007–2014
Walsh & Joshi (2024) [11]	Model selection by calibration vs. accuracy (NBA)	+34.7% vs. –35.2% avg. ROI
Wilkens (2021) [13]	Neural nets, random forests (tennis)	Accuracy capped ≈70%; no clear edge over odds
Wunderlich & Memmert (2020) [12]	Methodological	Betting returns are not a valid accuracy measure

SECTION 7

7. AI as guardian: integrity and player protection

AI does not only price markets; it also polices them. Sportradar’s Universal Fraud Detection System monitored more than one million events in 2025 and flagged 1,116 suspicious matches; the number identified by its AI systems rose by 56%, from 436 to 682 [45]. The International Betting Integrity Association reported 300 suspicious betting alerts in 2025, up 29% [46]. Rising alert numbers do not necessarily mean more corruption – they largely reflect better detection.

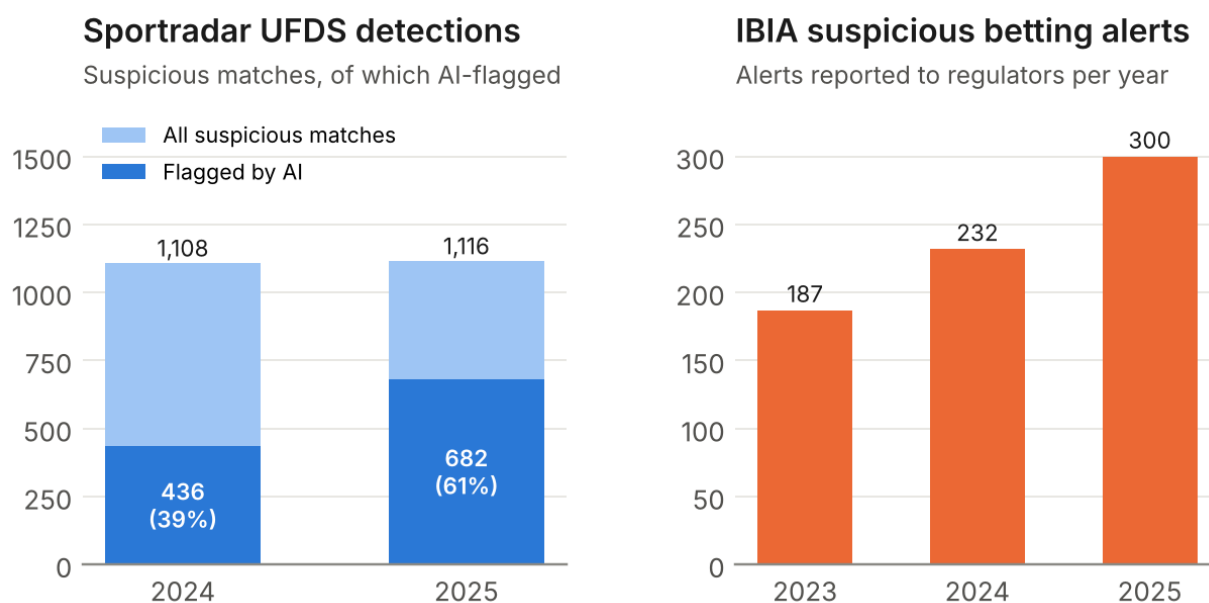


Figure 9. Left: suspicious matches detected by Sportradar UFDS, of which flagged by AI. Right: IBIA suspicious betting alerts. Sources: Sportradar [45], IBIA [46].

In player protection, regulation increasingly requires automation. Since September 2022, UK remote operators must act on “strong indicators of harm” using automated processes [43]. Specialist vendors such as Mindway AI now monitor around 15 million players per month [44]. In Germany, the GGL’s central LUGAS system enforces a cross-operator deposit limit of €1,000 per month and prevents parallel play, processing data from more than 60 licensed operators and about 5 million players [52]. The same

behavioural models that identify sharp bettors can identify vulnerable ones; which use they are prioritised for is a regulatory question, not a technical one.

SECTION 8

8. The German perspective: a margin with a tax on top

Germany adds a unique layer to the margin debate. Since July 2021, a 5.3% tax is levied on every sports bet stake (previously 5% since 2012). Bookmakers either pass it on through a deduction from stakes or winnings, or build it into lower odds. Sports betting tax receipts were €409 million in 2023 [49], consistent with legal stakes of around €7.7–7.9 billion; the German Sports Betting Association (DSWV), relaying GGL data, reports €8.2 billion for 2024 [47].

Original analysis: the effective cost of a Premier League bet in Germany

An average 2024/25 Premier League 1X2 market carries a 4.65% overround, i.e. a theoretical payout of $100 \div 104.65 = 95.6\%$ [23].

If the 5.3% stake tax is fully passed on to the bettor, the payout falls to $95.6\% \times (1 - 0.053) \approx 90.5\%$. The effective cost of the bet roughly doubles, from 4.4% to 9.5% of the stake.

On a five-leg bet builder at 5% per leg, the same calculation yields a payout of $78.4\% \times 0.947 \approx 74\%$. A German bettor on such a product gives up around a quarter of their stake on average.

The DSWV itself states an average payout ratio of around 85% for the German market and has rejected a 93% assumption used in academic estimates as unrealistic “given the 5.3% tax on each bet” [48]. France, with a legal payout cap of 85%, shows implied payout ratios of about 82–83% in recent years [50]. Both are far below the 95–97% theoretical payout of sharp single-market prices.

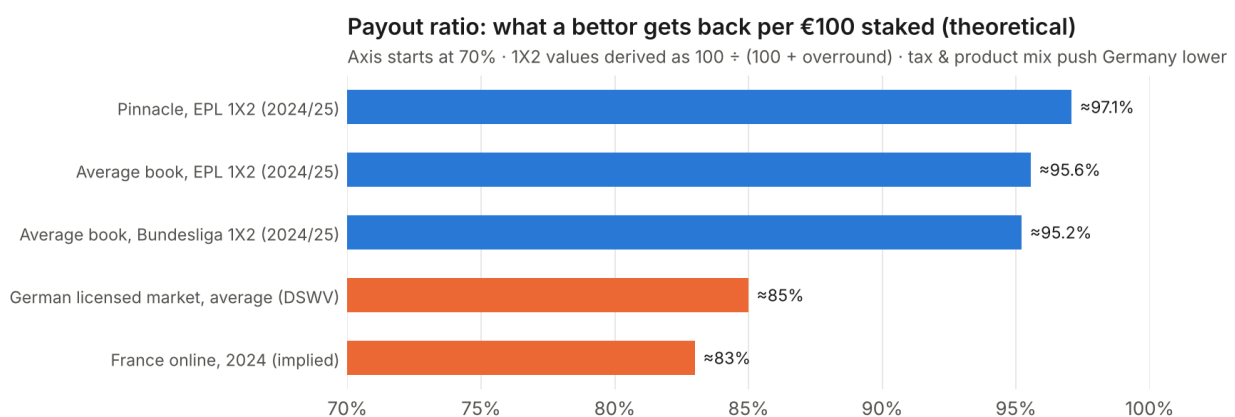


Figure 10. Theoretical payout ratio on major 1X2 markets compared with average payout ratios in the German and French regulated markets. Sources: Eastwood [23], DSWV via Meyer & Hayer / iGB [48], ANJ [50].

For the German market, the policy implication is significant. A turnover tax that is independent of margin hits low-margin single bets hardest in relative terms and narrows the gap between a sharp Premier League price and a high-margin combination bet. It thereby weakens one of the strongest arguments for staying in the licensed market – fair prices – at a time when the DSWV estimates the unlicensed share of the online market at around 25% [47]. A tax on gross gaming revenue, as used in most European markets, would let the price advantage of efficient single markets reach the customer.

SECTION 9

9. Outlook 2027–2030 and implications

Based on the evidence, I expect the Margin Paradox to deepen rather than resolve over the next five years:

- **Single-market margins stay flat to slightly rising.** The 4–5% level on top-league 1X2 markets appears to be a floor. Further compression requires a competitive shock – a new low-margin entrant or an exchange revival – rather than better technology.
- **Blended hold keeps rising.** FanDuel's 16% hold target for 2030 [34] shows where operators see the ceiling. Generative AI interfaces that let customers build bets in natural language (“Kane to score and Bayern to win by two”) will push the share of bespoke, high-margin combinations further.
- **Personalised pricing is the next frontier.** The data to offer different prices to different customers already exists through stake factoring. Explicit personalised odds would be the logical – and regulatorily contested – next step.
- **Transparency regulation will follow.** Massachusetts and the UK Gambling Commission are moving toward explaining account restrictions. Mandatory display of margins on combination bets would be a natural next measure.

Implications

Table 7. Practical implications of the Margin Paradox.

Stakeholder	What it means
Bettors	Price matters more than picks. Prefer single bets, Asian handicap and low-margin books; treat bet builders as entertainment with a 20–40% cost; judge models by calibration and closing-line value, not by short-term profit.
Operators	Margin is now managed through product mix and customer segmentation rather than headline odds. Long-term trust and regulatory licence depend on how transparently this is done.
Regulators	Monitor hold by product, not only GGR. Require disclosure of margins on combination products and of restriction criteria. In Germany, re-evaluate the stake-based tax.
Investors & suppliers	Proprietary pricing for correlated and live markets is the core asset. Suppliers that own data rights and AI pricing capture rising value; turnkey platforms face in-sourcing pressure.

FAQ

10. Frequently asked questions

Has AI made sports betting odds better for bettors?

On single, liquid markets such as the 1X2 market in top football leagues, yes – margins are less than half of what they were around 2000. But the average bettor today pays more per euro staked, because AI has enabled high-margin products such as bet builders, same-game parlays and micro-bets that now account for a large share of stakes.

What is a typical bookmaker margin in 2025?

About 4.5–5% on Premier League and Bundesliga 1X2 markets at the average bookmaker, around 3% at Pinnacle, about 15% on US bet builders and 20–35% on UK bet builders.

Why is US sportsbook hold rising?

Mainly because of parlays and same-game parlays. In New Jersey, parlays generated almost 60% of 2024 revenue on under 30% of handle. Excluding parlays, New Jersey's hold was about 4.9%.

Do bookmakers limit winning bettors?

Yes. UK Gambling Commission data for 2024 show that 4.31% of accounts were restricted, and 46.8% of restricted customers were in profit versus 25.4% of all customers.

Can machine learning beat the bookmaker?

Occasionally, in research settings and niche markets, particularly with well-calibrated models. But closing odds remain the best public forecast, and profitable accounts are quickly limited.

How does the 5.3% German betting tax affect odds?

If passed on fully, it reduces a typical 95.6% payout on a Premier League 1X2 bet to about 90.5%, roughly doubling the effective cost of the bet.

METHODOLOGY

11. Methodology and limitations

This paper is a structured synthesis of published evidence. Sources were identified in October 2026 through systematic searches of academic literature (International Journal of Forecasting, Journal of Sports Economics, Economic Journal, Machine Learning and others), regulatory publications (UK Gambling Commission, Massachusetts Gaming Commission, New Jersey DGE, Illinois Gaming Board, GGL, ANJ), industry associations (AGA, IBIA, DSWV) and company filings (Sportradar, Genius Sports, DraftKings, Flutter, Kambi).

Limitations should be noted. Historical margin figures stem from different studies with different bookmaker samples and are therefore shown as separate points rather than a continuous series. Several figures were taken from secondary summaries of the underlying studies and reports; readers relying on individual numbers should consult the primary sources listed. Vendor statements (e.g. on AI performance) are marked as claims. Calculations labelled as the author's own are simple derivations from the cited figures. No academic study yet links AI adoption causally to margin changes; the Margin Paradox framework is an interpretation of the combined evidence and should be tested with bookmaker-level data.

References

- [1] Forrest, D., Goddard, J., & Simmons, R. (2005). Odds-setters as forecasters: The case of English football. *International Journal of Forecasting*, 21(3), 551–564.
- [2] Štrumbelj, E. (2014). On determining probability forecasts from betting odds. *International Journal of Forecasting*, 30(4), 934–943.
- [3] Angelini, G., & De Angelis, L. (2019). Efficiency of online football betting markets. *International Journal of Forecasting*, 35(2), 712–721. doi:10.1016/j.ijforecast.2018.07.008
- [4] Elaad, G., Reade, J. J., & Singleton, C. (2020). Information, prices and efficiency in an online betting market. *Finance Research Letters*, 35, 101291. doi:10.1016/j.frl.2019.09.006
- [5] Winkelmann, D., Ötting, M., Deutscher, C., & Makarewicz, T. (2024). Are betting markets inefficient? Evidence from simulations and real data. *Journal of Sports Economics*, 25(1), 54–97.
- [6] Deutscher, C., Winkelmann, D., & Ötting, M. (2021). Bookmakers' mispricing of the disappeared home advantage in the German Bundesliga after the COVID-19 break. *Applied Economics*, 53(26), 3054–3064.
- [7] Croxson, K., & Reade, J. J. (2014). Information and efficiency: Goal arrival in soccer betting. *The Economic Journal*, 124(575), 62–91.

- [8] Spann, M., & Skiera, B. (2009). Sports forecasting: A comparison of the forecast accuracy of prediction markets, betting odds and tipsters. *Journal of Forecasting*, 28(1), 55–72.
- [9] Kaunitz, L., Zhong, S., & Kreiner, J. (2017). Beating the bookies with their own numbers – and how the online sports betting market is rigged. arXiv:1710.02824.
- [10] Hubáček, O., Šourek, G., & Železný, F. (2019). Exploiting sports-betting market using machine learning. *International Journal of Forecasting*, 35(2), 783–796; and Learning to predict soccer results from relational data with gradient boosted trees. *Machine Learning*, 108, 29–47.
- [11] Walsh, C., & Joshi, A. (2024). Machine learning for sports betting: Should model selection be based on accuracy or calibration? *Machine Learning with Applications*, 16, 100539.
- [12] Wunderlich, F., & Memmert, D. (2020). Are betting returns a useful measure of accuracy in (sports) forecasting? *International Journal of Forecasting*, 36(2), 713–722.
- [13] Wilkens, S. (2021). Sports prediction and betting models in the machine learning age: The case of tennis. *Journal of Sports Analytics*, 7(2).
- [14] Gomez-Gonzalez, C., & del Corral, J. (2018). The betting market over time: Overround and surebets in European football. *Economics and Business Letters*, 7(4), 129–136.
- [15] Che, X., Feddersen, A., & Humphreys, B. R. (2017). Price setting and competition in fixed odds betting markets. In *The Economics of Sports Betting* (pp. 38–51). Edward Elgar.
- [16] Hegarty, T., & Whelan, K. (2025). Forecasting soccer matches with betting odds: A tale of two markets. *International Journal of Forecasting*, 41(2), 803–820.
- [17] Whelan, K., & Hegarty, T. Estimating expected loss rates in betting markets. CEPR Discussion Paper DP17948 (published in *Applied Economics*, 2025); historical margin figures as cited therein.
- [18] Whelan, K. (2026). Compounding an edge? Expected utility and the puzzle of parlay betting. Working paper, University College Dublin.
- [19] Shin, H. S. (1993). Measuring the incidence of insider trading in a market for state-contingent claims. *The Economic Journal*, 103(420), 1141–1153.
- [20] Levitt, S. D. (2004). Why are gambling markets organised so differently from financial markets? *The Economic Journal*, 114(495), 223–246.
- [21] Kuypers, T. (2000). Information and efficiency: An empirical study of a fixed odds betting market. *Applied Economics*, 32(11), 1353–1363.
- [22] Reade, J. J., Singleton, C., & Vaughan Williams, L. (2019). Betting markets for English Premier League results and scorelines. University of Reading, Economics Discussion Paper 2019-10.
- [23] Eastwood, M. (2025, July 16). How accurate are soccer odds? A data dive into 250 million betting lines. pena.lt.
- [24] Mathematics of bookmaking (summary of a six-bookmaker study of English league margins, 2005/06–2017/18). [Wikipedia](https://en.wikipedia.org/wiki/Mathematics_of_bookmaking).
- [25] Newall, P. W. S. et al. (2017). Bookmaker margins in football betting, incl. 2014 FIFA World Cup product margins. [University of Stirling](https://www.universityofstirling.ac.uk/research/centres-and-institutes/centre-for-sport-and-recreation/research-projects/football-betting/).
- [26] Glush, E. (2026). Bookmaker Margin Panel [dataset]. Mendeley Data. doi:10.17632/yt3xsvk7zn.1; and Gaming & Co / Propus Partners (2023), French bookmakers: overround and margin analysis.
- [27] SBC News (2024, Jan 17). Market make-up: UK bookies battle on margin control of bet builders (Propus Partners data). [sbcsnews.co.uk](https://www.sbcnews.co.uk).
- [28] SBC News (2024, Jan 25). Olivier Kaplan: From Loto Foot to bet builder. [sbcsnews.co.uk](https://www.sbcnews.co.uk).
- [29] American Gaming Association: annual commercial gaming revenue reports 2019–2025, via SBC Americas, iGaming Business and ESPN. americangaming.org.
- [30] Legal Sports Report (2023–2026). Analyses of US sports betting hold. legalsportsreport.com.
- [31] New Jersey Division of Gaming Enforcement (2025). December 2024 gaming revenue press release. [nj.gov](https://www.nj.gov).
- [32] Illinois Gaming Board monthly sports wagering reports, via InGame and Action Network (2024–2026). [ingame.com](https://www.ingame.com).
- [33] DraftKings Inc. quarterly earnings calls 2023–2026, via Motley Fool, Legal Sports Report and MarketBeat. [marketbeat.com](https://www.marketbeat.com).
- [34] Flutter Entertainment Investor Day (2024) and Q1 2025 results, via Legal Sports Report and SBC Americas. legalsportsreport.com.
- [35] Kambi Group plc. Q4 2024 and Q4 2025 reports. [Q4 2024 report](https://www.kambi.com).
- [36] Sportradar Group AG. FY2024 results (SEC Ex. 99.1, March 2025) and FY2025 results (March 2026). [sportradar.com](https://www.sportradar.com).
- [37] European Gaming (2025, Apr 24). Sportradar expands Alpha Odds into cricket. europeangaming.eu.
- [38] Genius Sports Ltd. FY2025 results, SEC Ex. 99.1 (March 2026). [sec.gov](https://www.sec.gov).

- [39] DraftKings Inc. Annual Report on Form 10-K for FY2024 (Simplebet acquisition). [sec.gov](#).
- [40] SBC Americas (2025, Jan 17). In-play betting report (Optimove). [sbcamericas.com](#).
- [41] UK Gambling Commission (2025, Jul 23). Commercial restrictions by betting operators. [gamblingcommission.gov.uk](#).
- [42] Massachusetts Gaming Commission (2025). Meeting materials on sports wagering account limits, Sept–Dec 2025. [massgaming.com](#).
- [43] UK Gambling Commission (2022). Customer interaction guidance for remote gambling licensees. [gamblingcommission.gov.uk](#).
- [44] Mindway AI. Company information and CDC Gaming (2025). [mindway.ai](#).
- [45] Sportradar (2026, Feb 10). Sports integrity strengthens as global match-fixing declines in 2025. [sportradar.com](#).
- [46] International Betting Integrity Association (2025–2026). Annual integrity reports. [ibia.bet](#).
- [47] Gambling Insider (2025). GGL Q2 results: German sports stakes down 13.5%; DSWV market data. [gamblinginsider.com](#).
- [48] Meyer, G., & Hayer, T. (2025). Stellungnahme zum Deutschen Sportwettenverband; and iGaming Business, DSWV criticises DHS methodology. [igamingbusiness.com](#).
- [49] Destatis via dpa / Stuttgarter Zeitung (2024). Steuereinnahmen aus Glücksspiel sinken. [stuttgarter-zeitung.de](#).
- [50] Autorité nationale des jeux (2025). Bilan du marché des jeux d’argent et de hasard en France – 2024. [anj.fr](#).
- [51] H2 Gambling Capital for IBIA (2024). Global regulated sports betting market report. [h2gc.com](#).
- [52] SBC News (2026, Jul 23). LUGAS and the GGL interstate system. [sbcnews.co.uk](#).
- [53] Pinnacle. Pinnacle’s margins (Betting Resources). [pinnacle.com](#).

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